

Requirement Catalogue

Comparative Evaluation of Vision-Based Finger Tracking for Piano Interaction

1. Purpose and Scope

This document catalogues all functional and non-functional requirements defined for the Piano Finger Tracking data collection and analysis system. Requirements were established in the Initial Project Plan and are reproduced here in full with their MoSCoW priority and final completion status.

The system is a research prototype, not a consumer product. Its purpose is to benchmark MediaPipe Hands and OpenPose Hand models on their ability to identify which finger strikes which piano key, under controlled variations of skin tone, lighting, and hand size.

INFO MoSCoW key: Must = critical to project success. Should = high value but not blocking. Could = desirable if time permits. Won't = explicitly out of scope for this iteration.

2. MoSCoW Priority Summary

Priority	Total	Completed	Outcome
Must	15	14	14/15 completed
Should	4	4	All 4 Should requirements completed
Could	3	1	1/3 completed (2 descoped)
Won't	4	N/A	All 4 correctly excluded from scope

3. Functional Requirements

Functional requirements define what the system must do. Nineteen functional requirements were identified, covering data capture, model integration, calibration, analysis, and reporting.

ID	Requirement	MoSCoW	Status	Notes
FR-01	Finger landmark tracking using vision models	Must	Completed	Real-time detection of all 21 hand landmarks for both hands using at least one open-source model.

FR-02	MIDI ground truth capture (Casio CTK-2400)	Must	Completed	Capture MIDI note_on events via USB with sub-millisecond timestamp precision.
FR-03	Temporal synchronisation between MIDI and video	Must	Completed	Align MIDI events and video frames to a shared perf_counter origin set at recording start.
FR-04	Support multiple open-source models	Must	Partially completed	MediaPipe Hands and OpenPose Hand integrated. MoveNet excluded due to incompatible output format.
FR-05	Camera-to-keyboard spatial calibration	Must	Completed	Interactive homography-based polygon mask that warps key boundaries to match the camera's oblique view.
FR-06	Data logging system	Must	Completed	Per-session output: frames.csv (timestamps), MIDI JSONL, standard MIDI file, and session metadata JSON.
FR-07	Keypress event analysis	Must	Completed	Polygon containment test determines which key polygon a fingertip falls inside at each note_on event.
FR-08	Quantitative performance metrics	Must	Completed	MPJPE (horizontal-only), detection rate, and accuracy percentage computed per session and in aggregate.
FR-09	Controlled variation of experimental conditions	Must	Completed	Fitzpatrick skin type (II-V), lux category (Dim/Indoor), and hand size (7.5-12.5 cm) systematically logged.
FR-10	Statistical comparison of models	Must	Completed	Descriptive statistics (mean, SD) with grouped bar charts comparing MediaPipe and OpenPose outcomes.
FR-11	Model selection justification	Must	Completed	Literature review and methodology chapter justify model choices with reference to prior work.
FR-12	Reproducible experimental protocol	Should	Completed	Session protocol, hardware configuration, and analysis scripts documented and released.
FR-13	Evaluation under additional models	Could	Not completed	MoveNet and other models considered but excluded due to incompatible landmark output format.
FR-14	Real-time visualisation overlay	Should	Completed	live_preview.py shows hand skeleton, key polygons, MPJPE, and detection stats in real time.
FR-15	Automatic occlusion event annotation	Could	Not completed	Distinguishing true occlusion from model failure was descoped; counted as detection_fail.
FR-16	3D hand pose estimation	Won't	Correctly excluded	Single overhead camera cannot reconstruct 3D depth reliably; 2D horizontal MPJPE is sufficient.
FR-17	Mobile / embedded deployment	Won't	Correctly	Research prototype; real-time

			excluded	laptop deployment meets all performance requirements.
FR-18	Multi-camera / multi-view tracking	Won't	Correctly excluded	Single fixed-overhead camera is the defined hardware configuration for this study.
FR-19	Real-time audio feedback (gamification)	Won't	Correctly excluded	Out of research scope; project focuses on measurement, not interactive music education.

3.1 Must Requirements — Detail

FR-01 Finger landmark tracking using vision models

Real-time detection of all 21 hand landmarks for both hands using at least one open-source model.

FR-02 MIDI ground truth capture (Casio CTK-2400)

Capture MIDI note_on events via USB with sub-millisecond timestamp precision.

FR-03 Temporal synchronisation between MIDI and video

Align MIDI events and video frames to a shared perf_counter origin set at recording start.

FR-04 Support multiple open-source models

MediaPipe Hands and OpenPose Hand integrated. MoveNet excluded due to incompatible output format.

FR-05 Camera-to-keyboard spatial calibration

Interactive homography-based polygon mask that warps key boundaries to match the camera's oblique view.

FR-06 Data logging system

Per-session output: frames.csv (timestamps), MIDI JSONL, standard MIDI file, and session metadata JSON.

FR-07 Keypress event analysis

Polygon containment test determines which key polygon a fingertip falls inside at each note_on event.

FR-08 Quantitative performance metrics

MPJPE (horizontal-only), detection rate, and accuracy percentage computed per session and in aggregate.

FR-09 Controlled variation of experimental conditions

Fitzpatrick skin type (II-V), lux category (Dim/Indoor), and hand size (7.5-12.5 cm) systematically logged.

FR-10 Statistical comparison of models

Descriptive statistics (mean, SD) with grouped bar charts comparing MediaPipe and OpenPose outcomes.

FR-11 Model selection justification

Literature review and methodology chapter justify model choices with reference to prior work.

3.2 Should / Could / Won't Requirements — Detail

FR-12 Reproducible experimental protocol

Session protocol, hardware configuration, and analysis scripts documented and released.

FR-13 Evaluation under additional models

MoveNet and other models considered but excluded due to incompatible landmark output format.

FR-14 Real-time visualisation overlay

live_preview.py shows hand skeleton, key polygons, MPJPE, and detection stats in real time.

FR-15 Automatic occlusion event annotation

Distinguishing true occlusion from model failure was descoped; counted as detection_fail.

FR-16 3D hand pose estimation

Single overhead camera cannot reconstruct 3D depth reliably; 2D horizontal MPJPE is sufficient.

FR-17 Mobile / embedded deployment

Research prototype; real-time laptop deployment meets all performance requirements.

FR-18 Multi-camera / multi-view tracking

Single fixed-overhead camera is the defined hardware configuration for this study.

FR-19 Real-time audio feedback (gamification)

Out of research scope; project focuses on measurement, not interactive music education.

4. Non-Functional Requirements

Non-functional requirements define how the system should perform — covering quality attributes such as performance, ethics, reproducibility, and technology constraints.

ID	Requirement	MoSCoW	Status	Notes
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NFR-01	Research-focused functional prototype	Must	Completed	System functions as a data collection and analysis pipeline for academic research, not a consumer product.
NFR-02	Fair and controlled comparison conditions	Must	Completed	Identical hardware, camera position, and session protocol applied across all participants and both models.
NFR-03	Python + open-source libraries only	Must	Completed	Python 3.10, OpenCV, MediaPipe, Mido, NumPy, Pandas, Matplotlib. No proprietary SDKs.
NFR-04	Ethical compliance and participant protection	Must	Completed	University ethics approval (Stages 1 and 2), signed informed consent forms, anonymised participant IDs.
NFR-05	Real-time performance $\geq 15\text{--}20$ FPS	Should	Completed	Sustained 30 FPS on the host machine (MSI laptop, Intel i7-14th gen) during live_preview.py recording.
NFR-06	Code quality, documentation, reproducibility	Should	Completed	Organised script package, inline docstrings, INSTALLATION.md, User Guidelines, and this Requirement Catalogue.
NFR-07	Basic robustness and error handling	Could	Completed	Daemon-thread MIDI capture continues across hardware hiccups; graceful degradation when no MIDI port found.

4.1 Non-Functional Requirements — Detail

NFR-01 Research-focused functional prototype

System functions as a data collection and analysis pipeline for academic research, not a consumer product.

NFR-02 Fair and controlled comparison conditions

Identical hardware, camera position, and session protocol applied across all participants and both models.

NFR-03 Python + open-source libraries only

Python 3.10, OpenCV, MediaPipe, Mido, NumPy, Pandas, Matplotlib. No proprietary SDKs.

NFR-04 Ethical compliance and participant protection

University ethics approval (Stages 1 and 2), signed informed consent forms, anonymised participant IDs.

NFR-05 Real-time performance $\geq 15\text{--}20$ FPS

Sustained 30 FPS on the host machine (MSI laptop, Intel i7-14th gen) during live_preview.py recording.

NFR-06 Code quality, documentation, reproducibility

Organised script package, inline docstrings, INSTALLATION.md, User Guidelines, and this Requirement Catalogue.

NFR-07 Basic robustness and error handling

Daemon-thread MIDI capture continues across hardware hiccups; graceful degradation when no MIDI port found.

5. Requirements Traceability

The table below maps each Must and Should requirement to the script or system component that implements it.

Requirement ID	Implemented in	Verified by
FR-01	model_manager.py / live_preview.py	Live overlay + analyse.py output
FR-02	midi_recorder.MidiRecorder	p###_midi.jsonl timestamps
FR-03	midi_recorder.py (shared t0)	Frame/MIDI delta < 1 frame verified
FR-04	model_manager.MediaPipeHandModel + OpenPoseHandModel	Both models run on all sessions
FR-05	key_calibration.KeyMask (homography)	key_centers.json polygon vertices
FR-06	record.py / midi_recorder.py	data/raw/p###/ file listing
FR-07	analyse.py (pointPolygonTest)	matched / detection_fail counts
FR-08	analyse.py + plot_results.py	data/processed/ JSON + charts
FR-09	fitzpatrick_detector.py / lux_calculator.py	p###_session.json fields
FR-10	plot_results.py (bar charts, group means)	data/plots/ PNG files
FR-11	Report sections 3, 6, 8	Documented in main report
FR-12	INSTALLATION.md / User Guidelines.docx	Released in repository
FR-14	live_preview.py	Visual overlay + results panel
NFR-01	All scripts (no UI beyond OpenCV windows)	Prototype scope maintained
NFR-02	record.py setup screen (identical protocol)	Session metadata consistency
NFR-03	requirements.txt + piano_env/	All imports from open-source packages
NFR-04	Ethics docs / consent forms	Appendices C and D of main report
NFR-05	stats_collector.py (FPS overlay)	30 FPS sustained in live_preview.py
NFR-06	scripts/ package + documentation files	Code review + docs suite
NFR-07	midi_recorder.py (daemon thread)	MIDI capture continuity across errors

6. Explicitly Out-of-Scope Items

The following items were considered during project planning and deliberately excluded. They are recorded here to prevent scope creep in future iterations.

FR-16 3D hand pose estimation

Single overhead camera cannot reconstruct 3D depth reliably; 2D horizontal MPJPE is sufficient.

FR-17 Mobile / embedded deployment

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FR-18 Multi-camera / multi-view tracking

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NOTE These Won't items represent deliberate design constraints, not failures. They may be revisited in future work if hardware or scope changes.